Status	Finished
Started	Monday, 26 October 2026, 12:31 AM
Completed	Monday, 26 October 2026, 12:38 AM
Duration	7 mins 34 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Binary Search

b. Merge Sort

c. Bubble Sort

d. Linear Search

The correct answer is: Merge Sort

Question **2**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. sort()
- b. arrange()
- c. order()
- d. sorted()

The correct answer is: sorted()

Question **3**

Complete

Mark 0.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Quick Sort
- b. Bubble Sort
- c. Linear Search
- d. Merge Sort

The correct answer is: Merge Sort

Question **4**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort is less efficient for large lists
b. Bubble Sort is difficult to implement
c. Bubble Sort does not guarantee sorted order
d. Bubble Sort cannot handle duplicate elements

The correct answer is: Bubble Sort is less efficient for large lists

Question **5**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are split again into smaller sublists
b. They are compared element by element
c. They are discarded
d. They are combined to form a single sorted list

The correct answer is: They are combined to form a single sorted list

Question **6**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

a. Bubble Sort
b. Insertion Sort
c. Merge Sort
d. Quick Sort

The correct answer is: Bubble Sort

Question **7**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It makes the data unsorted
b. It complicates the selection of items
c. It makes it easier to select items based on their relationship to the rest of the items
d. It slows down the process

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **8**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Merging

b. Sorting

c. Searching

d. Complexity

The correct answer is:
Complexity

Question **9**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

a. A sorting algorithm is stable if it preserves the order of non-duplicate keys

b. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys

c. A sorting algorithm is stable if it preserves the order of duplicate keys

d. A sorting algorithm is stable if it preserves the order of all keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **10**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

- a. It is based on the divide-and-conquer approach
- b. It does not require recursion
- c. It is less efficient than bubble sort
- d. It sorts data using a single pass

The correct answer is: It is based on the divide-and-conquer approach

Question **11**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

- a. Dynamic programming
- b. Divide and conquer
- c. Backtracking
- d. Greedy method

The correct answer is:
Divide and conquer

Question **12**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Makes data harder to manage
b. Makes code execution slower
c. Decreases the efficiency of algorithms
d. Provides a basis for other algorithms to work efficiently

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **13**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It sorts data out of the box efficiently
b. It only works with integer arrays
c. It requires external libraries
d. It is less efficient than custom sorting algorithms

The correct answer is: It sorts data out of the box efficiently

Question **14**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Deleting data from a list
- b. Arranging data in a particular format
- c. Inserting data into a list
- d. Searching for data in a list

The correct answer is: Arranging data in a particular format

Question **15**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Linear Search
- b. Binary Search
- c. Merge Sort
- d. Bubble Sort

The correct answer is: Merge Sort